

Project: Inverse Eigenvalue Problem for Persymmetric Matrices with Prescribed Pattern

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A matrix $A = [a_{i,j}] \in \mathcal{M}_n(\mathbb{R})$ is called *persymmetric* if its entries satisfy the relation:

$$a_{i,j} = a_{n-j+1, n-i+1} \quad \text{for all } 1 \leq i, j \leq n.$$

Hence, persymmetric matrices are symmetric with respect to the anti-diagonal. Alternatively, let J denote the $n \times n$ matrix with ones on the anti-diagonal and zeros elsewhere. A matrix A is persymmetric if and only if $A = JA^\top J$. Let

$$J = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \text{ and } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 2 \\ 6 & 4 & 1 \end{pmatrix},$$

and note that A is persymmetric. Persymmetric matrices play an important role in signal processing and radar detection, and have applications in communications, information theory, and coding theory.

Let $\mathcal{P} = [p_{i,j}]$ be an $n \times n$ pattern matrix where each entry $p_{i,j}$ belongs to a set $\{+, -, 0\}$, indicating whether an entry is positive (+), negative (-), or equal to zero (0). A pattern $\mathcal{P}$ is said to be *persymmetric consistent* if it respects the persymmetric structure:

$$p_{i,j} = p_{n-j+1, n-i+1} \quad \text{for all } 1 \leq i, j \leq n.$$

By $\text{Per}\mathcal{S}(\mathcal{P})$ we denote the family of persymmetric matrices whose entries are constrained by the pattern $\mathcal{P}$. For example, for $\mathcal{P} = \begin{pmatrix} 0 & + & - \\ - & 0 & + \\ + & - & 0 \end{pmatrix}$ the set $\text{Per}\mathcal{S}(\mathcal{P})$ consists of all matrices of the form $\begin{pmatrix} 0 & p_1 & n_2 \\ n_1 & 0 & p_1 \\ p_2 & n_1 & 0 \end{pmatrix}$, where $p_i, i = 1, 2$, is positive, and $n_i, i = 1, 2$, is negative.

The Inverse Eigenvalue Problem (IEP) for $\text{Per}\mathcal{S}(\mathcal{P})$: Given a persymmetric consistent pattern $\mathcal{P}$ we ask which multisets of eigenvalues can be realised by matrices in the class $\text{Per}\mathcal{S}(\mathcal{P})$.

Illustrative aims that can be selected for this project are listed below:

- resolve the question for small dimensions,
- resolve the question for special cases of patterns $\mathcal{P}$,
- consider subproblems to the IEP (minimal rank, multiplicity lists),
- develop a strong spectral property for $\text{Per}\mathcal{S}(\mathcal{P})$,
- investigate patterns $\mathcal{P}$ with a nilpotent matrix in $\text{Per}\mathcal{S}(\mathcal{P})$,
- investigate spectrally arbitrary patterns.

Although researchers have studied the IEP for non-negative persymmetric matrices, the IEP for persymmetric matrices with prescribed pattern has not been investigated to date.